

Learning with Unknown Input Observers for Robust Nonlinear Estimation

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Abstract: This work introduces a robust hybrid state estimation framework for vehicle motion tracking that combines physics-based guarantees with data-driven adaptability. Standard model-based observers often fail when strict rank conditions are not met, while pure neural network approaches lack stability guarantees. To address this, we propose a two-stage architecture. First, we develop a Generalized Unknown Input Observer (UIO) that utilizes output derivatives to relax the standard rank condition, ensuring the existence of a bounded estimation of unmodeled dynamics. Second, we introduce a Neural Adaptive Observer that treats the UIO estimates as a "teacher" signal. Unlike the UIO, which provides instantaneous signal estimation, the neural observer learns the underlying structure of the uncertainty as a function of the state. This allows the system to predict disturbances along reference trajectories and refine the estimation error asymptotically. The approach is validated via Linear Matrix Inequalities (LMIs) ensuring $\mathcal{H}^1$ and $\mathcal{L}_2$ stability.

Keywords: Robust estimation , Model Learning , Neural network.

1. INTRODUCTION

State estimation is a cornerstone of autonomous vehicle control, where safety depends on accurate knowledge of system states and external disturbances. Traditional methods rely on explicit physics-based models (Barfoot, 2017). However, in vehicle dynamics, highly nonlinear phenomena-such as variable tire-road friction or aerodynamic drag are difficult to model mathematically, leading to estimation errors that compromise control performance.

Data-driven methods, particularly Neural Networks (NNs), have emerged as powerful tools for approximating these complex nonlinear functions (Abdollahi et al., 2006; Buisson-Fenet et al., 2021). While NNs offer superior adaptability, they pose significant risks in safety-critical applications. In online learning scenarios, small shifts in input data can cause NNs to behave unpredictably or diverge, lacking the stability guarantees provided by control theory. Consequently, recent research has shifted toward hybrid architectures that embed learning within observer frameworks.

A major limitation in existing hybrid observers is their reliance on the standard observer matching condition. For many vehicle systems, this condition does not hold, making it theoretically impossible to decouple the unknown input from the state using standard techniques. While some methods attempt to bypass this using approximations, they often result in unbounded errors or require expensive additional sensors.

To overcome these limitations, this paper proposes a layered Teacher-Student estimation architecture consisting of two distinct observers:

- A Generalized UIO (The Teacher): We introduce a regularization technique using output derivatives that ensures the observer exists even when the rank condition fails. This observer provides a robust, physics-guaranteed signal of the unknown input, ensuring the estimation error remains bounded.
- A Neural Adaptive Observer (The Student): While the UIO estimates the current value of the disturbance, it cannot predict how that disturbance behaves at future states (e.g., along a reference trajectory). The Neural Observer utilizes the UIO's signal to learn the unknown dynamics as a function of the state ($\mu_\theta(x)$). This allows for refined precision and enables the controller to anticipate unmodeled dynamics.

This combination offers a specific advantage: the Generalized UIO guarantees the system never diverges (safety), while the Neural Network minimizes the residual error over time (performance). The contributions of this paper are:

- A Generalized UIO design that relaxes the rank condition using output derivatives ;
- A hybrid learning framework where the UIO acts as a supervisor, enforcing consistency in the neural training loop;
- LMI-based stability conditions that guarantee joint convergence of both the state and model parameters.

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2. PROBLEM FORMULATION AND MOTIVATION

In this paper, we consider the class of continuous-time nonlinear systems in state-space form described by the following equations:

$$\begin{aligned}\dot{x} &= \varphi(x, u) + B\mu_x \\ y &= Cx,\end{aligned}\tag{1}$$

where $x \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ is the state vector, $y \in \mathbb{R}^{n_y}$ is the output measurement, $u \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$ is the control input, and $\mu_x \in \mathbb{R}^{n_\mu}$ is an unknown nonlinear function that encapsulates the system uncertainty including both unmodeled dynamics and external disturbances. The matrices $C \in \mathbb{R}^{n_y \times n_x}$ and $B \in \mathbb{R}^{n_x \times n_\mu}$ are constant and known matrices. The function $\varphi : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^{n_x}$ represents the known nonlinearity of the system.

Remark 1. For simplicity, we consider systems with linear output. Otherwise, if we have nonlinear output $y = h(x)$, we introduce a new state $\eta(t)$, with $\dot{\eta} \triangleq A_\eta \eta + \gamma y$ and $\eta_0 = \eta(0)$ known. Then, a new system with the augmented vector $\zeta = \begin{bmatrix} \eta \\ x \end{bmatrix}$ may be considered, which has as linear output the vector η .

Now, let us introduce the following assumptions which are required for our unknown input observer design methods.

Assumption 1. The output vector y has dimension greater than or equal to the number of unknown terms μ_x ; equivalently, $n_\mu \leq n_y$.

Assumption 2. The nonlinear function $\varphi(\cdot)$ is γ_φ -Lipschitz continuous on $\mathcal{X}$, uniformly on $u \in \mathcal{U}$.

Assumption 3. The unknown nonlinearity $\mu_x(\cdot)$ is Lebesgue measurable and $\mathcal{L}_2$ -bounded.

The objective of this paper is to estimate the unmodeled dynamics μ_x . More specifically, we propose to estimate μ_x by combining an unknown input observer (UIO) with a neural network-based approach, leveraging gradient-based methods to better capture the unmodeled components of the system. The key idea is to incorporate the UIO-based estimation into the loss function of the neural network, thereby enhancing residual regularization and ensuring consistency with the UIO. This, in turn, improves the robustness and performance of the overall estimation scheme. Unfortunately, it is not possible to estimate the unknown input, μ_x , if the system fails to satisfy the following main necessary condition:

$$\text{rank}(CB) = \text{rank}(B).\tag{2}$$

In addition, to be able to directly construct a UIO for system (1) to estimate simultaneously x and μ_x , the constraint (2) is not sufficient. For this, we should be able to write the system under the form:

$$\begin{aligned}\mathbb{E}\dot{\zeta} &= \varphi_\zeta(\zeta, u) \\ y &= \mathcal{H}\zeta\end{aligned}\tag{3}$$

with

$$\text{rank}\left(\begin{bmatrix} \mathbb{E} \\ \mathcal{H} \end{bmatrix}\right) = n_x + n_\mu.\tag{4}$$

For further details on this requirement, the reader is referred to Chaouche et al. (2022); Trinh and Fernando (2011); Zemouche and Boutayeb (2009); Hassan et al. (2013); Trinh et al. (2008) and the references therein.

To address this challenge, various techniques have been proposed in the literature, each offering a different approach to the problem. Without aiming for exhaustiveness, these techniques can be summarized as follows. Nevertheless, the problem remains open, and new ideas or improvements are still possible, although often at the cost of additional assumptions:

- *Inverting the system dynamics:* The system state is first estimated, and the unknown inputs are then obtained by inverting the dynamics (Phanomchoeng and Rajamani, 2014; Benallouch et al., 2017; Chandabadi and Marquez, 2014). However, the presence of disturbances and nonlinearities in the output signal or in the dynamics can make the problem particularly challenging.
- *Deriving the output vector $y(t)$:* In some cases, the derivatives of the output measurements are required (Hou and Müller, 1992; Phanomchoeng and Rajamani, 2014). This leads to the construction of a pseudo-output vector y_{new} , which enables the simultaneous estimation of x and μ_x . For instance, in the linear case without disturbances, one obtains

$$y_{\text{new}} = C_{\text{new}}x + C_\mu\mu_x,$$

where $\text{rank}(C_\mu) = n_\mu$. However, this approach is not suitable for nonlinear disturbed systems, since differentiating $y(t)$ introduces additional variables arising from the derivatives of the disturbance vector, as well as new nonlinearities in the pseudo-output vector. Consequently, satisfying the rank condition (2) with the new matrices becomes very difficult.

- *Adding sensors:* In some cases, augmenting the system with additional sensors is a natural way to overcome the rank condition (2). This approach introduces a new measurement vector y_{new} , where the original output $y(t)$ is used to estimate the state $x(t)$, and the additional outputs y_{new} are exploited to estimate the unknown input, μ_x . For example, in Phanomchoeng and Rajamani (2014) (and references therein), four sensors (i.e., four output measurements) were employed to estimate only two unknown inputs. However, the presence of disturbances can significantly decrease estimation performance. Moreover, additional sensors may be very expensive or, in some cases, unavailable.
- *Proportional-Integral Observer and Null Dynamics:* Some works in the literature have employed a Proportional-Integral Observer (PIO) together with null dynamics imposed on the estimated unknown input (Söffker et al., 1995; Bakhshande and Söffker, 2015). However, this assumption does not reflect the true dynamics of the unknown input, leading to inaccurate estimation. Other approaches assume constant inputs (Jeon et al., 2020), i.e., $\dot{\mu}_x \equiv 0$, but this restriction is overly conservative. Recently, a method for discrete-time systems was proposed in Nguyen et al. (2024); however, this approach yields delayed estimates of both the system state and the unknown inputs, which is often impractical to design reliable controllers.

The limitations of the aforementioned methods motivated us to adopt a different approach and propose a simple strategy to estimate the system state and the unknown

input without relying on conservative assumptions. The key idea is to recognize that this initial estimation is not final. Rather, the objective is to provide a first-level approximation of x and μ_x , which will subsequently be refined through the neuro-adaptive observer we introduce later. This refinement enhances estimation accuracy and ultimately yields a definitive estimate of the unmodeled nonlinearity, μ_x .

The strategy consists in rewriting system (1) without modifying it. Then, the UIO we will propose will be based on reliable approximation of (1).

$$\begin{aligned} \dot{x} + E_{\delta_1} \dot{\mu}_x &= \varphi(x, u) + B\mu_x + E\omega_{\delta_1} \\ y &= Cx + D_{\delta_2} \mu_x + D\omega_{\delta_2} \end{aligned} \quad (5)$$

where $D \in \mathbb{R}^{n_y \times n_\mu}$ and $E \in \mathbb{R}^{n_x \times n_\mu}$ are given constant, full-column-rank matrix specified by the user; $E_{\delta_1} = \delta_1 E$, $D_{\delta_2} = \delta_2 D$; and $\omega_{\delta_1}(t) = \delta_1 \dot{\mu}_x(t)$, $\omega_{\delta_2}(t) = \delta_2 \mu_x(t)$, with $\delta_j \geq 0$, with $(\delta_1, \delta_2) \neq (0, 0)$, chosen sufficiently small to ensure the feasibility of some sufficient conditions that will be stated later. The purpose of this reformulation is to recover the rank condition (4), at the expense of introducing the additive terms $\omega_{\delta_j}(t)$, $j = 1, 2$, which are handled as disturbances.

The next section presents LMI conditions that guarantee accurate estimation of x and μ_x while satisfying an $\mathcal{H}_\infty$ criterion, or more specifically, an $\mathcal{L}_2$ -optimality criterion—where the performance bound is proportional to δ_j , $j = 1, 2$.

3. LMI-BASED $\mathcal{H}^1$ UIO DESIGN

This section is devoted to the numerical LMI-based design procedure ensuring the asymptotic estimation of the system state x and the unmodeled dynamics μ_x following a $\mathcal{H}^1$ -optimality criterion.

3.1 System transformation

Recall that $\mathcal{H}^1$ is the *Hilbert* space corresponding to the *Sobolev* space, $\mathcal{W}^{1,2}$, of square-integrable functions whose weak first derivatives are also square-integrable. Mathematically, we define the set $\mathcal{H}^1(\Omega)$ as

$$\mathcal{H}^1(\Omega) := \left\{ \psi \in \mathcal{L}^2(\Omega) \mid \frac{d\psi}{dt} \in \mathcal{L}^2(\Omega) \right\}$$

with $\Omega = [0, +\infty[$ in our case. This space is endowed by the norm $\|\cdot\|_{\mathcal{H}^1}$ defined as follows:

$$\|\psi\|_{\mathcal{H}^1} := \left(\|\psi\|_{\mathcal{L}^2}^2 + \left\| \frac{d\psi}{dt} \right\|_{\mathcal{L}^2}^2 \right)^{1/2}.$$

First, system (5) can be rewritten under the compact form:

$$\begin{aligned} \mathbb{E} \dot{\zeta} &= \varphi_\zeta(\zeta, u) + \bar{E} \omega_\delta \\ y &= \mathcal{H} \zeta + \bar{D} \omega_\delta \end{aligned} \quad (6)$$

where

$$\mathbb{E} := [\mathbb{I}_{n_x} \ E_{\delta_1}], \quad \mathcal{H} := [C \ D_{\delta_2}], \quad (7a)$$

$$\bar{E} := [E \ 0_{n_x \times n_\mu}], \quad \bar{D} := [0_{n_y \times n_\mu} \ D], \quad (7b)$$

$$\zeta := \begin{bmatrix} x \\ \mu_x \end{bmatrix}, \quad \omega_\delta := \begin{bmatrix} \omega_{\delta_1} \\ \omega_{\delta_2} \end{bmatrix}, \quad (7c)$$

$$\varphi_\zeta(\zeta, u) := \varphi(x, u) + B\mu_x. \quad (7d)$$

Now, let us introduce the following necessary assumption:

Assumption 4. The matrices E and D are chosen such that $\mathbb{E}$ and $\mathcal{H}$ satisfy the rank condition (4).

The objective is to design a state observer that provides an estimate $\hat{\zeta}$ of ζ , such that the following $\mathcal{H}^1$ -optimality inequality holds:

$$\|\tilde{\zeta}\|_{\mathcal{H}^1} \leq \lambda_\delta \|\mu_x\|_{\mathcal{H}^1} + \beta \|\xi_0\| \quad (8)$$

where $\tilde{\zeta} := \zeta - \hat{\zeta}$, λ_δ is the disturbance attenuation level, explicitly depending on δ_1 and δ_2 , $\beta > 0$ is a weighting real constant, and ξ_0 is a vector depending on $\tilde{\zeta}(0)$ and $\mu_x(0)$.

3.2 UIO structure and error dynamics

From Assumption 4, there exist two matrices P_ζ and Q_ζ such that

$$P_\zeta \mathbb{E} + Q_\zeta \mathcal{H} = \mathbb{I}_{n_\zeta}. \quad (9)$$

where $n_\zeta := n_x + n_\mu$. These matrices are exploited in the following observer structure:

$$\begin{cases} \dot{\eta} = P_\zeta \varphi_\zeta(\hat{\zeta}, u) + K(y - \mathcal{H}\hat{\zeta}) \\ \hat{\zeta} = \eta + Q_\zeta y \end{cases} \quad (10)$$

where $\hat{\zeta}$ is the estimate of ζ and $K \in \mathbb{R}^{n_\zeta \times \mathbb{R}^{n_y}}$ is the observer gain to be determined such that the estimation error, $\tilde{\zeta} := \zeta - \hat{\zeta}$, satisfies the $\mathcal{H}^1$ -optimality criterion (8) with appropriate λ_δ , β , and ξ_0 .

We have

$$\begin{aligned} \tilde{\zeta} &= \zeta - \eta - Q_\zeta y \\ &= (\mathbb{I}_{n_\zeta} - Q_\zeta \mathcal{H}) \zeta - \eta - Q_\zeta \bar{D} \omega_\delta \\ &\stackrel{(9)}{=} P_\zeta \mathbb{E} \zeta - \eta - Q_\zeta \bar{D} \omega_\delta \end{aligned} \quad (11)$$

which gives equivalently

$$\overbrace{\tilde{\zeta} + Q_\zeta \bar{D} \omega_\delta}^{\xi(t)} = P_\zeta \mathbb{E} \zeta(t) - \eta(t). \quad (12)$$

It is worth noting that the reformulation (12) conveniently avoids the derivative of ω_δ in the derivation of the error dynamics.

By using the dynamics (6) and (10), we obtain

$$\begin{aligned} \dot{\xi} &= P_\zeta \left[\varphi_\zeta(\zeta, u) - \varphi_\zeta(\hat{\zeta}, u) \right] - K \mathcal{H} \tilde{\zeta} \\ &\quad + (P_\zeta \bar{E} - K \bar{D}) \omega_\delta \\ &= \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K \mathcal{H} \right) \xi \\ &\quad + \left(\mathcal{E}(\mathbf{z}, u) - K \bar{D} \right) \omega_\delta \end{aligned} \quad (13)$$

where

$$\begin{aligned} \mathcal{E}(\mathbf{z}, u) &:= \bar{E} - P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) Q_\zeta \bar{D} \\ \mathcal{D} &:= (\mathbb{I}_{n_y} + \mathcal{H} Q_\zeta) \bar{D}. \end{aligned}$$

and from the differential mean value theorem (Arezki et al., 2023), we have

$$\varphi_\zeta(\zeta, u) - \varphi_\zeta(\hat{\zeta}, u) = \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) \tilde{\zeta}$$

for a given $\mathbf{z}$.

From this point onward, the analysis will be carried out with the vector ξ instead of $\tilde{\zeta}$, and we will return to $\tilde{\zeta}$ at the end to derive the final $\mathcal{H}^1$ bound (8).

3.3 New LMI-based UIO design conditions

Before presenting the proposed LMI-based UIO design procedure, we state an intermediate and general result in the following proposition.

Proposition 1. Let $\vartheta(\cdot)$ be a Lyapunov function such that there exists $\vartheta_{\max} > 0$ such that $\vartheta(z) \leq \vartheta_{\max} \|z\|^2$, for all $z \in \mathbb{R}^{n_\zeta}$. Consider a matrix $\mathcal{S} = \mathcal{S}^\top > 0$ and a real constant $\lambda > 0$ such that the following inequality holds:

$$\dot{\vartheta}(z) + \|z\|^2 + \dot{z}^\top \mathcal{S} \dot{z} - \lambda \|w\|^2 \leq 0, \quad \forall z \in \mathbb{R}^{n_\zeta}, w \in \mathbb{R}^{2n_\mu} \quad (14)$$

with $z \in \mathcal{H}^1$ and $w \in \mathcal{L}^2$. Then, the following inequality is satisfied:

$$\|z\|_{\mathcal{H}^1} \leq \sqrt{\frac{\lambda}{\min(1, \lambda_{\min}(\mathcal{S}))}} \|w\|_{\mathcal{L}^2} + \sqrt{\frac{\vartheta_{\max}}{\min(1, \lambda_{\min}(\mathcal{S}))}} \|z_0\|. \quad (15)$$

Moreover, if (14) is satisfied with $z := \xi$ and $w := \omega_\delta$, where ξ and ω_δ are defined by (12) and (7c), respectively, then the estimation error, $\tilde{\zeta}$, satisfies the $\mathcal{H}^1$ bound (8) with λ_δ , β , and ξ_0 given as follows:

$$\lambda_\delta := \delta_2 \sigma_{\max}(Q_\zeta D) + \max(\delta_1, \delta_2) \sqrt{\frac{\lambda}{\min(1, \lambda_{\min}(\mathcal{S}))}} \quad (16a)$$

$$\beta := \sqrt{\frac{\vartheta_{\max}}{\min(1, \lambda_{\min}(\mathcal{S}))}} \quad (16b)$$

$$\xi_0 := \tilde{\zeta}(0) + Q_\zeta D \mu_x(0). \quad (16c)$$

where $\sigma_{\max}(Q_\zeta D)$ represents the largest singular value of the matrix $Q_\zeta D$.

Proof. The first part of the proof is relatively straightforward. Before integrating (14) to get the norms $\|\cdot\|_{\mathcal{H}^1}$ and $\|\cdot\|_{\mathcal{L}^2}$, take in mind that

$$\min(1, \lambda_{\min}(\mathcal{S})) (\|z\|^2 + \|\dot{z}\|^2) \leq \|z\|^2 + \dot{z}^\top \mathcal{S} \dot{z}.$$

Hence, since

$$\int_0^{+\infty} (\|z(s)\|^2 + \|\dot{z}(s)\|^2) ds = \|z\|_{\mathcal{H}^1}^2,$$

$$\int_0^{+\infty} \|w(s)\|^2 ds = \|w\|_{\mathcal{L}^2}^2,$$

and because $\vartheta(z(t)) \geq 0$ for all $t \geq 0$, implying that

$$\int_0^{+\infty} \vartheta(z(s)) ds \geq -\vartheta(z(0)),$$

we can readily conclude (15).

As for the second part of the proof, we will exploit the structure of ξ and ω_δ . First, ξ and ω_δ satisfy (15) proved in the first part. In addition, we have

$$\begin{aligned} \|\tilde{\zeta}\|_{\mathcal{H}^1} &= \|\xi - Q_\zeta \bar{D} \omega_\delta\|_{\mathcal{H}^1} \\ &\leq \|\xi\|_{\mathcal{H}^1} + \|Q_\zeta \bar{D} \omega_\delta\|_{\mathcal{H}^1} \\ &= \|\xi\|_{\mathcal{H}^1} + \|[0 \ Q_\zeta D] \omega_\delta\|_{\mathcal{H}^1} \\ &= \|\xi\|_{\mathcal{H}^1} + \delta_2 \|Q_\zeta D \mu_x\|_{\mathcal{H}^1} \\ &\leq \|\xi\|_{\mathcal{H}^1} + \delta_2 \sigma_{\max}(Q_\zeta D) \|\mu_x\|_{\mathcal{H}^1} \end{aligned} \quad (17)$$

and

$$\begin{aligned} \|\omega_\delta\|_{\mathcal{L}^2} &= \sqrt{\delta_1^2 \|\dot{\mu}_x\|_{\mathcal{L}^2}^2 + \delta_2^2 \|\mu_x\|_{\mathcal{L}^2}^2} \\ &\leq \max(\delta_1, \delta_2) \sqrt{\|\dot{\mu}_x\|_{\mathcal{L}^2}^2 + \|\mu_x\|_{\mathcal{L}^2}^2} \\ &= \max(\delta_1, \delta_2) \|\mu_x\|_{\mathcal{H}^1}. \end{aligned} \quad (18)$$

Hence, by substituting (17) and (18) in (15), the bound (8) is inferred with the parameters given in (16a)–(16c).

Before stating the main theorem, notice that from Assumption 2, there exist constant matrices $\mathcal{A}_j \in \mathbb{R}^{n_\zeta \times n_\zeta}$ and functions $\alpha_j(z)$, $j = 1, \dots, \bar{n}_\zeta$ such that the Jacobian $\nabla_{\zeta}^{\varphi_\zeta}(z, u)$ belongs to the convex polytopic set defined as:

$$\mathcal{H}_\varphi \triangleq \left\{ \sum_{j=1}^{\bar{n}_\zeta} \alpha_j(z) \mathcal{A}_j, \sum_{j=1}^{\bar{n}_\zeta} \alpha_j(z) = 1, \alpha_j(z) \geq 0 \right\} \quad (19)$$

where $\mathcal{A}_j \in \mathbb{R}^{n_\zeta \times n_\zeta}$, are the vertices of the polytope $\mathcal{H}_\varphi$.

Now, we are ready to state the main theorem, which provides new LMI conditions ensuring the bound (8).

Theorem 1. Suppose that Assumptions 1, 2, and 3 hold. Let the matrices E and D , together with the scalars δ_1 and δ_2 , be chosen such that the Assumption 4 is satisfied. For a given constant $\epsilon > 0$, assume further that there exist a symmetric positive definite matrix $\mathcal{P} \in \mathbb{R}^{n_\zeta \times n_\zeta}$ and a matrix $\mathcal{Q} \in \mathbb{R}^{n_y \times n_\zeta}$ such that the LMI conditions (26) are fulfilled. Then, the estimation error $\tilde{\zeta}$, with $K = \mathcal{P}^{-1} \mathcal{Q}^\top$, satisfies the $\mathcal{H}^1$ -optimality bound (8) with the parameters given in (16a)–(16c), where $\vartheta_{\max} = \lambda_{\max}(\mathcal{P})$ and $\mathcal{S} = \epsilon \mathcal{P}$ with $\lambda_{\min}(\mathcal{S}) = \epsilon \lambda_{\min}(\mathcal{P})$.

Proof. Consider the quadratic Lyapunov function $\vartheta(\xi) := \xi^\top \mathcal{P} \xi$. Then, the derivative of $\vartheta(\cdot)$ along the trajectories of (13) is given as follows:

$$\begin{aligned} \dot{\vartheta}(\xi) &= \xi^\top \left[\left(P_\zeta \nabla_{\zeta}^{\varphi_\zeta}(z, u) - K \mathcal{H} \right)^\top \mathcal{P} \right. \\ &\quad \left. + \mathcal{P} \left(P_\zeta \nabla_{\zeta}^{\varphi_\zeta}(z, u) - K \mathcal{H} \right) \right] \xi \\ &\quad + \xi^\top \mathcal{P} \left(\mathcal{E}(z, u) - K \mathcal{D} \right) \omega_\delta \\ &\quad + \omega_\delta^\top \left(\mathcal{E}(z, u) - K \mathcal{D} \right)^\top \mathcal{P} \xi. \end{aligned} \quad (20)$$

and

$$\begin{aligned} \xi^\top \mathcal{S} \xi &= \xi^\top \left(P_\zeta \nabla_{\zeta}^{\varphi_\zeta}(z, u) - K \mathcal{H} \right)^\top \mathcal{S} \times \\ &\quad \left(P_\zeta \nabla_{\zeta}^{\varphi_\zeta}(z, u) - K \mathcal{H} \right) \xi \\ &\quad + \omega_\delta^\top \left(\mathcal{E}(z, u) - K \mathcal{D} \right)^\top \mathcal{S} \left(\mathcal{E}(z, u) - K \mathcal{D} \right) \omega_\delta \\ &\quad + 2 \xi^\top \left(P_\zeta \nabla_{\zeta}^{\varphi_\zeta}(z, u) - K \mathcal{H} \right)^\top \mathcal{S} \left(\mathcal{E}(z, u) - K \mathcal{D} \right) \omega_\delta. \end{aligned} \quad (21)$$

Then, we have

$$\dot{\vartheta}(\xi) + \|\xi\|^2 + \xi^\top \mathcal{S} \xi - \lambda \|\omega_\delta\|^2 = \begin{bmatrix} \xi \\ \omega_\delta \end{bmatrix}^\top \mathbb{M}(z, u) \begin{bmatrix} \xi \\ \omega_\delta \end{bmatrix} \quad (22)$$

where $\mathbb{M}(z, u)$ is defined in (24). By applying Schur lemma, we deduce that $\mathbb{M}(z, u) < 0$ whenever inequality (25) holds. Consequently, by setting $\mathcal{S} = \epsilon \mathcal{P}$, introducing the change of variables $\mathcal{Q} := K^\top \mathcal{P}$, and invoking the convexity principle, we conclude that the inequality $\mathbb{M}(z, u) < 0$ is satisfied provided that the LMI conditions (26) hold. This, in turn, yields

$$\dot{\vartheta}(\xi) + \|\xi\|^2 + \xi^\top \mathcal{S} \dot{\xi} - \lambda \|\omega_\delta\|^2 \leq 0. \quad (23)$$

Hence, Proposition 1 applies and the desired result follows.

Remark 2. It is worth emphasizing that the choice $\mathcal{S} = \epsilon \mathcal{P}$ is neither arbitrary nor overly restrictive, although it represents a specific selection. The introduction of the matrix $\mathcal{P}$ serves to recover the decision variable $\mathcal{Q}$ and then to avoid bilinear matrix inequalities, which are generally unsuitable for numerical solvers. The scalar parameter ϵ is included as a *feasibility factor* to improve the feasibility of the LMI conditions (26), and it should be chosen a priori.

4. OUTPUT DERIVATIVE-BASED GENERALIZED UIO

The main limitation of the previous result is that the proposed method does not guarantee exponential convergence of the estimation error, even when condition (2) is satisfied. This represents a drawback of the proposed *regularization* technique and has motivated the development of a more general and robust method that overcomes this shortcoming.

4.1 Output derivative-based generalized model

The key idea consists in introducing a generalized system imbedding the original model, the output y and its derivative. To this end, let us consider $\chi_1 \triangleq y$ and $\chi_2 \triangleq \dot{y}$, which leads to the following generalized system:

$$\begin{cases} \dot{\chi}_1 = \chi_2 \\ \dot{\chi}_2 = C \bar{\varphi}_x(x, u) + C \frac{\partial \varphi}{\partial x}(x, u) B \mu_x + C B \dot{\mu}_x \\ \dot{x} = \varphi(x, u) + B \mu_x \\ y_\chi = \chi_1 + C_\chi \chi_2 + C x \end{cases} \quad (27)$$

where $\bar{\varphi}_x(x, u) \triangleq \frac{\partial \varphi}{\partial x}(x, u) \varphi(x, u)$, and $C_\chi \in \mathbb{R}^{n_y \times n_y}$ is a given weighting matrix associated with the measured derivative $\dot{y}$.

Clearly, if (2) is satisfied, then (27) can be rewritten in the form of (6), for which Assumption 4 holds. In this case, an exponential UIO for (27) can be directly constructed if we add χ_2 as output measurement. However, if (2) is not satisfied, the construction of an exponential UIO is not possible. In this situation, we can proceed as in the previous section by first regularizing the system (27). This approach leads to a design method that is more general than that of the previous section, as it ensures exponential convergence of the UIO whenever condition (2) is satisfied and the derivative $\dot{y}$ is used as a measurement.

4.2 Regularization of the generalized model

System (27) can be regularized as follows

$$\begin{cases} \dot{\chi}_1 = \chi_2 \\ \dot{\chi}_2 + (E_{\delta_1} - C B) \dot{\mu}_x = C \bar{\varphi}_x(x, u) + E \omega_{\delta_1} + C \frac{\partial \varphi}{\partial x}(x, u) B \mu_x \\ \dot{x} = \varphi(x, u) + B \mu_x \\ y_\chi = \chi_1 + C_\chi \chi_2 + C x + D_{\delta_2} \mu_x + D \omega_{\delta_2} \end{cases} \quad (28)$$

where the scalars $\delta_1 \geq 0$, $\delta_2 \geq 0$, and the matrices $E \in \mathbb{R}^{n_y \times n_\mu}$ and $D \in \mathbb{R}^{n_y \times n_\mu}$ are the *regularization parameters*, with E and D chosen to be binary, i.e., their entries are restricted to $\{0, 1\}$. This choice is motivated by the aim of controlling the estimation error bound exclusively through the scalar parameters δ_1 and δ_2 . Moreover, E and D are selected to satisfy an appropriate rank condition, which will be specified later.

First, system (5) can be rewritten under the compact form:

$$\begin{aligned} \mathbb{E} \dot{\zeta} &= \varphi_\zeta(\zeta, u) + \bar{E} \omega_\delta \\ y_\chi &= \mathcal{H} \zeta + \bar{D} \omega_\delta \end{aligned} \quad (29)$$

where

$$\mathbb{E} := \begin{bmatrix} \mathbb{I}_{n_y} & 0 & 0 & 0 \\ 0 & \mathbb{I}_{n_y} & 0 & -C B \\ 0 & 0 & \mathbb{I}_{n_x} & 0 \end{bmatrix}, \quad \mathcal{H} := [\mathbb{I}_{n_y} \quad C_\chi \quad C \quad D_{\delta_2}], \quad (30a)$$

$$\bar{E} := \begin{bmatrix} 0 & 0_{n_y \times n_\mu} \\ E & 0_{n_y \times n_\mu} \\ 0 & 0_{n_x \times n_\mu} \end{bmatrix}, \quad \bar{D} := [0_{n_y \times n_\mu} \quad D], \quad (30b)$$

$$\zeta := \begin{bmatrix} \chi_1 \\ \chi_2 \\ x \\ \mu_x \end{bmatrix}, \quad \omega_\delta := \begin{bmatrix} \omega_{\delta_1} \\ \omega_{\delta_2} \end{bmatrix}, \quad (30c)$$

$$\varphi_\zeta(\zeta, u) := \begin{bmatrix} \chi_2 \\ C \bar{\varphi}_x(x, u) + C \frac{\partial \varphi}{\partial x}(x, u) B \mu_x \\ \varphi(x, u) + B \mu_x \end{bmatrix} \quad (30d)$$

with E_{δ_1} , D_{δ_2} , ω_{δ_1} , and ω_{δ_2} defined as in (5).

4.3 Generalized UIO

Here we provide the generalized UIO corresponding to the generalized and regularized system (29).

Assumption 5. The matrices E , D , and C_χ , and the scalars δ_1 and δ_2 are chosen such that $\mathbb{E}$ and $\mathcal{H}$ satisfy the rank condition (4).

Assumption 6. The function $\varphi_\zeta(\cdot, u)$ defined in (30d) is globally Lipschitz, uniformly on u .

As in the previous section, from Assumption 5, there exist two matrices P_ζ and Q_ζ such that

$$P_\zeta \mathbb{E} + Q_\zeta \mathcal{H} = \mathbb{I}_{n_\zeta}. \quad (31)$$

where $n_\zeta := 2n_y + n_x + n_\mu$, and then an observer is constructed as in (10) as follows:

$$\begin{cases} \dot{\hat{\eta}} = P_\zeta \varphi_\zeta(\hat{\zeta}, u) + K (y_\chi - \mathcal{H} \hat{\zeta}) \\ \dot{\hat{\zeta}} = \hat{\eta} + Q_\zeta y_\chi \end{cases} \quad (32)$$

$$\begin{bmatrix} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right)^\top \mathcal{P} + \mathcal{P} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right) & \left[\mathcal{P} + \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right)^\top \mathcal{S} \right] \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right) \\ \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right)^\top \left[\mathcal{P} + \mathcal{S} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right) \right] & -\lambda \mathbb{I}_{2n_\mu} + \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right)^\top \mathcal{S} \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right) \end{bmatrix} \quad (24)$$

$$\begin{bmatrix} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right)^\top \mathcal{P} + \mathcal{P} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right) & \mathcal{P} \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right) & \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right)^\top \mathcal{S} \\ \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right)^\top \mathcal{P} & -\lambda \mathbb{I}_{2n_\mu} & \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right)^\top \mathcal{S} \\ \mathcal{S} \left(P_\zeta \nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u) - K\mathcal{H} \right) & \mathcal{S} \left(\mathcal{E}(\mathbf{z}, u) - K\mathcal{D} \right) & -\mathcal{S} \end{bmatrix} < 0 \quad (25)$$

$$\begin{bmatrix} \left(P_\zeta \mathcal{A}_j \right)^\top \mathcal{P} + \mathcal{P} \left(P_\zeta \mathcal{A}_j \right) - \mathcal{H}^\top \mathcal{Q} - \mathcal{Q}^\top \mathcal{H} & \mathcal{P} \mathcal{E}_j - \mathcal{Q}^\top \mathcal{D} & \left(P_\zeta \mathcal{A}_j \right)^\top \mathcal{P} - \mathcal{H}^\top \mathcal{Q} \\ \mathcal{E}_j^\top \mathcal{P} - \mathcal{D}^\top \mathcal{Q} & -\lambda \mathbb{I}_{2n_\mu} & \mathcal{E}_j^\top \mathcal{P} - \mathcal{D}^\top \mathcal{Q} \\ \mathcal{P} \left(P_\zeta \mathcal{A}_j \right) - \mathcal{Q}^\top \mathcal{H} & \mathcal{P} \mathcal{E}_j - \mathcal{Q}^\top \mathcal{D} & -\frac{1}{\epsilon} \mathcal{P} \end{bmatrix} < 0 \quad (26)$$

where

$$\mathcal{E}_j := \bar{E} - P_\zeta \mathcal{A}_j Q_\zeta \bar{D}$$

To avoid repetitions, under Assumption 6, assume that the Jacobian $\nabla_\zeta^{\varphi_\zeta}(\mathbf{z}, u)$ belongs to the convex polytopic set $\mathcal{H}_\varphi$ defined in (19).

It is unnecessary to restate a new theorem or proposition here. Since we use the same notation as in Section 3, the results of Proposition 1 and Theorem 1 apply directly under Assumptions 5 and 6. Specifically, the observer (32) with $K = \mathcal{P}^{-1} \mathcal{Q}^\top$ satisfies the $\mathcal{H}^1$ -optimality bound (8), with parameters given in (16a)–(16c), where $\vartheta_{\max} = \lambda_{\max}(\mathcal{P})$ and $\mathcal{S} = \epsilon \mathcal{P}$, with $\lambda_{\min}(\mathcal{S}) = \epsilon \lambda_{\min}(\mathcal{P})$. The matrices $\mathcal{P}$, $\mathcal{Q}$, and the scalar ϵ are defined in Theorem 1.

4.4 Specific cases and discussion

This section is devoted to discussing the general result established in Section 4. In addition, several specific cases are analyzed to illustrate the benefits of the generalized UIO in Section 4.3 relative to that of Section 3.

Case $\text{rank}(CB) = \text{rank}(B)$ We show that in this case, exponential convergence of the estimation error to zero can be guaranteed. Indeed, if $\text{rank}(CB) = \text{rank}(B)$, then to satisfy Assumption 5, it suffices to choose an appropriate matrix C_χ , independently of δ_1 and δ_2 . We can then set $\delta_1 = \delta_2 = 0$, which implies $\lambda_\delta = 0$ in (16a) and $\omega_\delta(t) \equiv 0$ in (23). Under these conditions, the bound (8) reduces to

$$\|\tilde{\zeta}\|_{\mathcal{H}^1} \leq \beta \|\xi_0\|, \quad (33)$$

which, combined with the uniform continuity of $\tilde{\zeta}(\cdot)$, implies

$$\lim_{t \rightarrow +\infty} \tilde{\zeta}(t) = 0.$$

In other words, we have $\xi(t) = \tilde{\zeta}(t)$, and the LMIs (26) ensure (23), which reduces to

$$\dot{\vartheta}(\tilde{\zeta}(t)) + \|\tilde{\zeta}(t)\|^2 \leq 0,$$

thus guaranteeing the *exponential convergence* of $\tilde{\zeta}(t)$ to zero.

Case $C_\chi = 0$: In this case, the derivative of the output measurement, $\dot{y}$, is not used as an additional input to the observer. Unfortunately, even when $\text{rank}(CB) = \text{rank}(B)$, the only way to satisfy Assumption 5 is to choose an appropriate nonzero matrix D and $\delta_2 > 0$. Consequently, we may set $\delta_1 = 0$ to achieve a tighter $\mathcal{H}^1$ bound; however, exponential convergence of the estimation error to zero cannot be guaranteed.

Case $\text{rank}(CB) < \text{rank}(B)$: In this case, exponential convergence of $\tilde{\zeta}$ to zero cannot be guaranteed; however, an $\mathcal{H}^1$ bound can still be obtained through regularization. By appropriately selecting the matrices E and D , and choosing the scalars δ_1 and δ_2 such that Assumption 5 is satisfied, the $\mathcal{H}^1$ bound (8) can be effectively adjusted by tuning the parameters δ_1 and δ_2 .

On the use of $\dot{y}$: The generalized UIO (32) requires real-time knowledge of the derivative $\dot{y}$, which can be viewed as a drawback since this derivative must be estimated online. Nevertheless, the structure of the proposed observer naturally incorporates the estimation of $\dot{y}$ through the auxiliary state variable χ_2 . In particular, when $\text{rank}(CB) = \text{rank}(B)$, the estimate $\hat{\chi}_2$ converges exponentially to χ_2 , thereby providing an accurate estimation of $\dot{y}$. This represents a major improvement over conventional UIO design approaches, which typically estimate the unknown inputs by differentiating the estimated states or the measured output y a posteriori.

5. UIO-BASED NEURAL ONLINE APPROXIMATION OF UNMODELED NONLINEARITY μ_x

As noted in the previous section, the approximation of the unmodeled nonlinearity μ_x is not yet final. We will employ neural network-based modeling to approximate it explicitly for control design purposes. The earlier UIO scheme serves only as a foundation for the online learning of μ_x . Regarding the estimation of the system state x , two approaches are possible: (1) use $\hat{x}$ from the existing UIO,

or (2) introduce a neuro-adaptive observer to simultaneously re-estimate x and the weighting parameters in the neural network representation of μ_x . In this paper, we will examine the first approach in detail.

First, we introduce the following assumption;

Assumption 7. The system can track the known trajectory x_{ref} under the observer-based controller as

$$u := \kappa(x_{\text{ref}}, u_{\text{ref}}, \hat{x}) \quad (34)$$

where $\hat{x}$ is the estimation of x and u_{ref} is a reference controller such that the pair $(x_{\text{ref}}, u_{\text{ref}})$ is an admissible trajectory of system (1).

5.1 Fully UIO-based learning

The generalized UIO derived in Section IV yields the augmented estimate $\hat{\zeta}$. To facilitate the proposed learning strategy, we decompose this vector to isolate the physics-based state estimate, denoted $\hat{x}_{\text{UIO}}$, and the unknown input estimate, $\hat{\mu}_x$. This partition is defined as:

$$\hat{\zeta} = \begin{bmatrix} \hat{\chi}_1 \\ \hat{\chi}_2 \\ \hat{x}_{\text{UIO}} \\ \hat{\mu}_x \end{bmatrix} \in \mathbb{R}^{2n_y + n_x + n_\mu} \quad (35)$$

Leveraging these physics-based priors, we introduce a neural network approximation $\mu_\theta(\cdot)$ to capture the residual unmodeled dynamics. This data-driven term is integrated into the observer structure to refine the estimation of μ_x online.

Let us consider $\mu_\theta(x)$, the neural network based approximation of $\hat{\mu}_x(t)$. In a general manner, we write

$$\mu_\theta(x) = \mathcal{N}_\theta(x, \hat{x}_{\text{UIO}}, \hat{\mu}_x, u) \quad (36)$$

where $\mathcal{N}_\theta(\cdot)$ represents the neural network function and θ is the neural network parameters. Then, this approximation is substituted into an observer to improve estimation, as follows:

$$\begin{aligned} \dot{\hat{x}}_{\text{NN}} &= \varphi(\hat{x}_{\text{NN}}, u) + B\mu_\theta(\hat{x}_{\text{NN}}) + L_{\text{NN}}(y - y_{\text{NN}}) \\ y_{\text{NN}} &= C\hat{x}_{\text{NN}} \end{aligned} \quad (37)$$

where $\hat{x}_{\text{NN}} \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ is the state of the observer (37) and y_{NN} is its output vector.

The proposed hybrid strategy is depicted in Figure 1. This framework combines a physics-based UIO, which exploits the measured output y to provide reliable initial estimates $\hat{x}_{\text{UIO}}$ and $\hat{\mu}_x$, with a data-driven Neural Observer. A neural network learns the unmodeled dynamics μ_θ to refine the state estimate $\hat{x}_{\text{NN}}$. This refined estimate is utilized by the controller for trajectory tracking, while the network parameters θ are updated online via a composite loss function that enforces consistency between the physics-based and data-driven models.

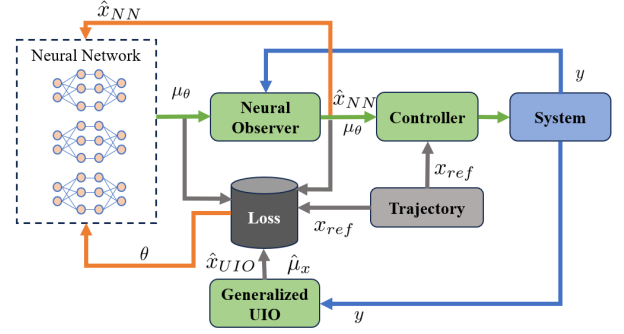


Fig. 1. Diagram of the proposed hybrid estimation strategy.

The original system (1) can be written under the form:

$$\dot{x} = \varphi(x, u) + B\mu_\theta(\hat{x}_{\text{NN}}) + B\Delta_x(t) \quad (38)$$

where $\Delta_x(t) \triangleq \mu_x(t) - \mu_\theta(\hat{x}_{\text{NN}}(t))$ represents the learning approximation error.

Design of the learning-based observer (37): The objective consists in determining the observer gain L_{NN} such that the estimation error, $\epsilon_{\text{NN}} \triangleq x - \hat{x}_{\text{NN}}$, satisfies the following $\mathcal{L}^2$ -optimality criterion:

$$\|\epsilon_{\text{NN}}\|_{\mathcal{L}^2} \leq \lambda_{\text{NN}} \|\Delta_x\|_{\mathcal{L}^2} + \beta_{\text{NN}} \|\epsilon_{\text{NN}}(0)\| \quad (39)$$

where λ_{NN} and β_{NN} are positive scalars. Towards this end, we employ the usual Lyapunov-based technique by using the quadratic Lyapunov function

$$\vartheta(\epsilon_{\text{NN}}) \triangleq \epsilon_{\text{NN}}^\top \mathcal{P} \epsilon_{\text{NN}}$$

where $\mathcal{P} = \mathcal{P}^\top > 0$. Then, to satisfy (39), we need to have

$$\dot{\vartheta}(\epsilon_{\text{NN}}) + \epsilon_{\text{NN}}^\top \epsilon_{\text{NN}} - \lambda_{\text{NN}}^2 \Delta_x^\top \Delta_x < 0.$$

First, the estimation error dynamic is expressed by the following equation:

$$\dot{\epsilon}_{\text{NN}} = \left(\nabla_x^\varphi(z_x, u) - L_{\text{NN}}C \right) \epsilon_{\text{NN}} + B\Delta_x(t) \quad (40)$$

where $\nabla_x^\varphi(z_x, u) = \varphi(x, u) - \varphi(\hat{x}_{\text{NN}}, u)$ for a given $z_x \in \mathbb{R}^{n_x}$, from the differential mean value theorem. According to Assumption 2, there exist constant matrices $\mathcal{A}_j^x \in \mathbb{R}^{n_x \times n_x}$ and functions $\alpha_j(z_x)$, $j = 1, \dots, \bar{n}_x$ such that the Jacobian $\nabla_x^\varphi(z_x, u)$ belongs to the convex polytopic set defined as:

$$\mathcal{H}_\varphi^x \triangleq \left\{ \sum_{j=1}^{\bar{n}_x} \alpha_j(z_x) \mathcal{A}_j^x, \sum_{j=1}^{\bar{n}_x} \alpha_j(z_x) = 1, \alpha_j(z_x) \geq 0 \right\} \quad (41)$$

where $\mathcal{A}_j^x \in \mathbb{R}^{n_x \times n_x}$, are the vertices of the polytope $\mathcal{H}_\varphi^x$.

Now, we are ready to state the sufficient LMI conditions guaranteeing the criterion (39).

Theorem 2. Under the Assumption 2, if there exist a symmetric positive definite matrix $\mathcal{P} \in \mathbb{R}^{n_x \times n_x}$, a matrix $\mathcal{Q} \in \mathbb{R}^{n_y \times n_x}$, and a positive scalar σ such that the following LMI conditions (42) are fulfilled:

$$\begin{bmatrix} (\mathcal{A}_j^x)^\top \mathcal{P} + \mathcal{P} \mathcal{A}_j^x - C^\top \mathcal{Q} - \mathcal{Q}^\top C & \mathcal{P} B \\ B^\top \mathcal{P} & -\sigma \mathbb{I}_{n_\mu} \end{bmatrix} < 0 \quad (42)$$

then, the estimation error ϵ_{NN} , with $L_{\text{NN}} = \mathcal{P}^{-1} \mathcal{Q}^\top$, satisfies the $\mathcal{L}^2$ -optimality criterion (39) with $\lambda_{\text{NN}} = \sqrt{\sigma}$ and $\beta = \sqrt{\lambda_{\text{max}}(\mathcal{P})}$.

Proof. The proof is omitted, as it is straightforward and well-established in the literature on $\mathcal{L}^2$ -optimality stability analysis.

Computation of the learning parameter θ : To approximate $\mu_x(t)$ via $\mu_\theta(\cdot)$, the parameter θ is optimized to minimize a specific loss function. Unlike classical supervised learning, where the ground truth $\mu_x(t)$ is accessible, we must rely on the UIO estimate $\hat{\mu}_x(t)$ as a proxy target (see Fig. 1). Consequently, we define a composite loss function that incorporates tracking errors and regularization terms to ensure robust convergence:

$$\begin{aligned} \mathcal{L}(\hat{x}_{\text{NN}}(\theta)) = & \underbrace{(\hat{x}_{\text{NN}} - x_{\text{ref}})^\top \mathcal{T}_{\text{ref}} (\hat{x}_{\text{NN}} - x_{\text{ref}})}_{\text{Tracking error}} \\ & + \underbrace{(y - y_{\text{NN}})^\top \mathcal{T}_y (y - y_{\text{NN}})}_{\text{Output error}} \\ & + \underbrace{(\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}})^\top \mathcal{T}_{\text{UIO}} (\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}})}_{\text{Consistency with UIO}} \\ & + \underbrace{\lambda \|\mu_\theta(\hat{x}_{\text{NN}}) - \hat{\mu}_x\|^2}_{\text{Residual regularization}} \end{aligned} \quad (43)$$

where $\mathcal{T}_{\text{ref}}$, $\mathcal{T}_y$, and $\mathcal{T}_{\text{UIO}}$ are symmetric and positive definite weight matrices of appropriate dimensions, and $\lambda > 0$ is a regularization gain.

Each quadratic term in the loss function (43) serves a specific purpose:

- *Tracking error term* $(\hat{x}_{\text{NN}} - x_{\text{ref}})^\top \mathcal{T}_{\text{ref}} (\hat{x}_{\text{NN}} - x_{\text{ref}})$: this term, retained from the original loss, penalizes deviations of the estimated state $\hat{x}_{\text{NN}}$ from the reference trajectory x_{ref} , ensuring the observer meets its primary objective of accurate tracking.
- *Output error term* $(y - y_{\text{NN}})^\top \mathcal{T}_y (y - y_{\text{NN}})$: by aligning the neural network predicted output y_{NN} with actual measurements y , this term provides direct supervision from sensor data. It stabilizes training, particularly in early stages, and enhances the observer ability to adapt to real-time conditions.
- *Consistency with UIO associated term* $(\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}})^\top \mathcal{T}_{\text{UIO}} (\hat{x}_{\text{NN}} - \hat{x}_{\text{UIO}})$: this term forces the neural network state estimate $\hat{x}_{\text{NN}}$ to remain consistent with the UIO-based estimate, leveraging the stability of the physics-based model. It facilitates a smooth transition from the UIO estimate to the neural network data-driven refinement, preventing significant divergence when the neural network is still learning.
- *Residual regularization term* $\lambda \|\mu_\theta(\hat{x}_{\text{NN}}) - \hat{\mu}_x\|^2$: this term regularizes the neural network learned dynamics $\mu_\theta(\hat{x}_{\text{NN}})$ to stay plausible relative to the estimate $\hat{\mu}_x$, mitigating overfitting and ensuring robustness.

6. CONCLUSION

This paper presented a robust hybrid estimation framework designed to address the challenges of vehicle motion tracking in the presence of unmodeled nonlinearities. We proposed a layered architecture that integrates a physics-based Generalized Unknown Input Observer (UIO) with a data-driven Neural Adaptive Observer. To overcome the structural limitations of standard observers, specifically when the rank condition $\text{rank}(CB) = \text{rank}(B)$ is not met, we introduced a regularization technique combined with

an output-derivative based generalized model. This approach ensures the existence of the observer and provides a reliable initial estimate of the unknown dynamics. This estimate subsequently serves as a supervisor for a neural network, which refines the approximation of complex nonlinearities, such as variable tire-road friction, through online learning. Theoretical analysis demonstrated that the proposed method guarantees $\mathcal{H}^1$ and $\mathcal{L}^2$ stability for the estimation errors, provided that the derived Linear Matrix Inequality (LMI) conditions are satisfied. Future work will focus on the experimental validation of this framework using real driving data and its extension to discrete-time formulations for embedded implementation.

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